Live GPS Tracker using ESP32 and NEO-6M GPS Module

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1. Introduction

With the rapid growth of vehicles and transportation services, tracking and monitoring the location of vehicles has become extremely important. Problems like vehicle theft, route deviation, driver misuse, and delayed deliveries are common in daily life. Parents also want to track school buses, companies want to monitor delivery vehicles, and owners want to protect personal vehicles.

Traditional security systems such as alarms and mechanical locks only help after the theft occurs and do not provide real-time information about the vehicle location. Therefore, a smarter and more reliable system is required.

A GPS Tracker System is an embedded system that continuously determines the vehicle's geographic location and sends it to the owner's mobile phone. The system uses satellite-based positioning along with a cellular communication network to provide real-time tracking.

This project proposes a low-cost, real-time GPS tracking device that can be installed inside any vehicle. The system will:

- Detect vehicle location using satellites
- Send coordinates to the user via GSM network

• Allow the owner to monitor the vehicle remotely

• Provide safety in case of theft or emergency

The goal is to build a reliable and affordable tracking solution suitable for bikes, cars, and delivery vehicles.

2. Literature Review

2.1 Existing Systems

Modern GPS trackers available in the market mainly rely on mobile applications and internet connectivity. They use GPS satellites to calculate location and upload the data to cloud servers.

Most tracking systems provide:

• Location tracking

• Route history

• Speed monitoring

However, these systems have some drawbacks:

- Expensive subscription charges
- Dependence on internet connectivity
- High power consumption
- Complex installation
- Many commercial trackers stop working in rural areas where internet connectivity is weak.

2.2 Gap Analysis

Our project removes the dependency on continuous internet and cloud servers. Instead, the system directly sends the location to the owner's phone using SMS, making it more reliable.

Gap Analysis Table

Feature	Commercial GPS Tracker Approach	Our Proposed GPS Tracker System (Context-Aware)
Main Control Method	App-based monitoring; user must open app to check location.	On-demand SMS tracking; system replies automatically when requested.
Tracking Method	Continuous periodic data upload to server.	User-request based tracking reduces unnecessary transmission.
Power Usage	High due to constant internet communication.	Low power using sleep mode and SMS communication.
Security/Authorization	Login credentials required in app.	Only registered mobile number receives location SMS.
The 'Human Factor'	User must continuously monitor app.	Works instantly whenever user sends command SMS.

2.3 Conclusion

Existing trackers focus on cloud connectivity and advanced dashboards but ignore affordability and reliability. Our project focuses on a simple, low-cost, and real-time solution that works even without mobile data, making it suitable for students and small vehicle owners.

3. Components Used

Components Used

Component	Specification/Example	Purpose in Project

Microcontroller Board	Arduino Uno / ATmega328P	Controls the entire system and processes GPS data
GPS Module	NEO-6M	Obtains latitude and longitude from satellites
GSM Module	SIM800L / SIM900	Sends SMS containing the location to the user
SIM Card	2G/3G enabled SIM	Provides cellular network connectivity
Power Supply	12V Vehicle Battery + 5V Regulator	Powers the tracker safely inside the vehicle
Voltage Regulator / Buck Converter	LM2596 or 7805	Converts vehicle voltage to stable 5V/4V
Antenna	External GSM/GPS Antenna	Improves signal reception
Connecting Wires	Jumper wires	Electrical connections between components
Enclosure	Plastic casing/box	Protects the circuit from dust and vibration

4. Working Principle

The GPS tracker works using two technologies:

1. GPS (Global Positioning System)

The GPS module communicates with satellites orbiting the Earth. It calculates:

- Latitude
- Longitude
- Speed
- Time

2. GSM Communication

The GSM module sends the location information to the owner's mobile phone using SMS.

Step-by-Step Operation:

- GPS module receives satellite signals
- Microcontroller reads coordinates
- User sends SMS command (like “TRACK”)
- System processes request
- GSM module replies with location link
- User opens Google Maps
- The received SMS contains a map link which directly opens the vehicle location.

5. System Architecture

Input: GPS satellite signals

Processing: Microcontroller

Output: SMS location to mobile phone

Flow:

GPS → Microcontroller → GSM → User Mobile

6. Relevant Technologies / Comparison

Technologies Used

Technology	Purpose in Project
Embedded Systems	Controls hardware components and processes sensor data via microcontroller.
GPS (Global Positioning System)	Determines real-time geographic coordinates (latitude & longitude) using satellites.
GSM Communication	Sends location information to user mobile phone via SMS.
Serial Communication (UART)	Allows communication between GPS module, GSM module, and microcontroller.
Power Management	Regulates vehicle battery voltage using regulator/buck converter for safe operation.
IoT Concept (Remote Monitoring)	Enables remote vehicle monitoring without physical presence.

Unlike IoT trackers, our system works without Wi-Fi or internet.

7. Relevant Patents

1. GPS Tracking and Monitoring System

Patent Number: US 8,355,850 B2
Assignee: AT&T Mobility II LLC
Grant Date: January 15, 2013

This patent describes a wireless GPS tracking device that determines the real-time location of a vehicle and transmits the data to a remote user through a cellular network. The system collects geographic coordinates using GPS satellites and then sends the information to a user interface such as a mobile phone or monitoring station.

Relevance to Our Project

This patent validates the core idea of our system — tracking a moving object using GPS positioning and transmitting the location through a communication network. Our project implements the same concept but in a simpler and low-cost embedded version.

Limitation of the Patent

The patented system depends on a large monitoring infrastructure and remote servers for processing and visualization of data.

Improvement in Our Project

Our project removes the dependency on external monitoring infrastructure. Instead of servers and dashboards, the system directly sends the location to the user via SMS. This reduces complexity, cost, and setup requirements.

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2. Vehicle Anti-Theft Tracking System

Patent Number: US 7,460,957 B1
Grant Date: December 2, 2008

This patent proposes a vehicle theft detection system that combines GPS and GSM communication to locate stolen vehicles. The device is hidden inside the vehicle and automatically sends alerts to the owner when suspicious movement is detected.

Relevance to Our Project

Our GPS tracker also focuses on vehicle safety and theft prevention. When the owner sends a request message, the system replies with the vehicle's current location, allowing recovery in case of theft.

Limitation

The system uses complex detection mechanisms and dedicated security infrastructure which increases cost.

Improvement in Our Project

Our design keeps the hardware minimal and affordable so that even a bike owner or student can use it. The focus is on accessibility and simplicity rather than expensive automation.

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3. Location Reporting System Using Mobile Communication

Patent Number: US 9,167,403 B2
Grant Date: October 20, 2015

This patent describes a portable tracking device that reports location periodically using mobile communication networks. It is commonly applied in fleet monitoring and logistics.

Relevance

This confirms the practical usefulness of GSM-based tracking systems in real-world transportation.

Improvement

Instead of periodic updates (which consume more power), our system works on **request-based tracking** — it only sends location when the user asks. This significantly reduces battery usage.

8. Product Case Study

Product Selected: Commercial GPS Vehicle Tracker (e.g., JioMotive / Onelap / typical GSM trackers)

Product Overview

Commercial GPS trackers are widely used in logistics companies, taxis, delivery services, and rental vehicles. These devices connect to GPS satellites to determine location and then upload the data to cloud servers through mobile internet (GPRS/4G). Users can view the vehicle location on a mobile app.

Typical features:
• Live tracking on mobile app
• Route history
• Speed monitoring
• Geo-fencing alerts

How the Commercial System Works

1. GPS module reads coordinates from satellites
2. Device sends data to cloud server via internet
3. Server processes the data
4. User accesses it through mobile application

Critical Analysis & Identified Gaps (Rectification)

Feature	Commercial GPS Tracker Solution	Our GPS Tracker System (Rectification)
Dependency	Cloud/app dependent; requires continuous internet connectivity for live tracking.	GSM SMS-based; works without internet and does not depend on any cloud server.
Cost & Complexity	High device cost, subscription fees, and professional installation required.	Low-cost, plug-and-play device with no subscription charges.

Target Audience	Fleet management companies, logistics operators, and large businesses.	Individual vehicle owners, students, and small vehicle users.

Key Observations

Commercial trackers are powerful but designed for fleet companies and businesses. They are expensive and depend heavily on internet connectivity and cloud servers. If the internet connection fails, live tracking stops.

Our project is designed for **individual users**, especially students and small vehicle owners. It provides essential tracking functionality without requiring applications, servers, or subscription fees.

Conclusion from Case Study

The case study shows that current market solutions focus on advanced features rather than accessibility. Our project simplifies the architecture by removing cloud dependency and using SMS-based communication.

This makes the system:
- Affordable
- Reliable
- Easy to install
- Suitable for rural and low-network areas

Hence, our GPS Tracker System acts as a lightweight and practical alternative to commercial tracking devices.

9. CORS Analysis

C - Challenges	1. GPS signal loss in tunnels, basements, or dense urban areas leading to inaccurate location. 2. GSM network availability issues or SMS delays in remote/rural regions. 3. Power management inside vehicle and protection from voltage spikes and noise.

O - Opportunities	1. Vehicle theft recovery and real-time owner awareness. 2. Fleet and delivery vehicle monitoring for businesses and logistics. 3. Safety monitoring for school buses and family vehicles.

R - Recommendations	1. Install the GPS antenna near the windshield and use an external antenna if required. 2. Use a DC-DC buck converter and fuse for safe power from the vehicle battery. 3. Provide a small rechargeable backup battery and tamper alert.

S - Solutions	1. Retry positioning and combine multiple satellite readings to improve accuracy. 2. On-demand SMS tracking and sleep mode to reduce power consumption. 3. Add geofencing alerts and periodic heartbeat messages to confirm system health.

10. Future Scope

The system can be enhanced by:

• Mobile application integration

• Live tracking on website

• Geo-fencing alerts

• Accident detection

• Engine immobilizer (remote vehicle stop)

• Cloud data storage

11. Descriptive Feature Analysis

- **Real-Time Tracking**

  The owner can instantly know the vehicle location through SMS.

- **Theft Protection**

  If the vehicle is stolen, the user can track it and inform police.

- **Emergency Use**

  Can be used in school buses, taxis, and transport vehicles.

- **Low Power Operation**

  The microcontroller uses sleep mode to reduce battery consumption.

12. Conclusion

The GPS Tracker System is a practical and cost-effective solution for vehicle monitoring and security. It provides real-time location tracking without requiring internet connectivity or expensive subscriptions. The system is simple to install, easy to operate, and highly useful for personal and commercial vehicles.

This project demonstrates how embedded systems and wireless communication can be combined to solve real-world problems such as vehicle theft and safety monitoring.